Large Language Model as a Teaching Assistant: Leveraging LLM for Multiple Choice Question Answering

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Attention Avengers

Abstract

Large language models (LLMs) have the potential to greatly benefit education by assisting students and teachers in their endeavor for a more efficient and enjoyable learning experience. With the increasing integration of automated systems in educational settings, there is a pressing need to optimize these models for specific tasks. This project aims to contribute to this goal by adapting the open-source LLM (0.5B Qwen1.5) for multiple-choice question answering (MCQ). To achieve this, a two-step training process was designed, beginning with Direct Preference Optimization (DPO) and followed by supervised fine-tuning for MCQ. Additionally, quantization was applied to reduce the memory needs of the model during inference. Although DPO did not yield significant improvement due to the quality of the dataset, the subsequent fine-tuning for MCQ demonstrated a notable increase in performance from the baseline on the MMLU dataset (37% vs 30% respectively). This study highlights the potential of targeted fine-tuning in enhancing LLM capabilities for educational applications.

1 Introduction

The rise of large language models (LLMs) has introduced a wealth of new opportunities in education. We have seen LLMs being applied to question solving (Wu et al., 2023), error correction (Zhang et al., 2023), confusion helping (Balse et al., 2023), automatic grading (Funayama et al., 2023) and content personalization (Yadav et al., 2023) among others. Although many of these approaches have not yet achieved market readiness, these advancements are set to transform the way students learn and interact with educational material.

In this project, we aim to leverage open-source LLM, specifically the 0.5B parameter chat version of Qwen1.5, for multiple-choice question answering with limited computational resources. We explore different fine-tuning techniques, experiment with generating training data, and investigate various quantization methods. More specifically, we begin with Direct Preference Optimization (DPO) (Rafailov et al., 2023) fine-tuning, which utilizes preference pairs generated by GPT-3.5. It is then followed by fine-tuning the model for multiple-choice questions and applying quantization for more efficient inference.

2 Related Work

Foundation models have revolutionized the field of natural language processing by providing powerful, pre-trained models capable of performing a wide range of tasks (Bommasani et al., 2022). These models, which are trained on vast amounts of data, serve as a good foundation for specific applications.

To adapt pre-trained, general-purpose LLMs to these specific tasks, various methods have been developed. One effective approach is prompt engineering, which involves carefully designing and optimizing input prompts to guide LLMs, ensuring that their responses are accurate and relevant (Chen et al., 2024). It is cost and computation efficient as it does not involve retraining the model and it has demonstrated its effectiveness in complex decision making tasks (Yao et al., 2023).

Fine-tuning the pre-trained model on smaller instruction datasets can also significantly enhance its performance in various tasks. While traditional reinforcement learning techniques involve training a reward model to identify and reward desirable LLM outputs (Christiano et al., 2023), recent advances like direct preference optimization (DPO) compare two outputs and indicate a preference, allowing the LLM to learn and adjust its behavior without needing a reward model (Rafailov et al., 2023). Several other variations of DPO have recently emerged that account for some noise in the preference pairs. By, for example, tweaking the loss function, these methods aim to generate more

robust training results (Mitchell, 2023; Chowdhury et al., 2024).

Fine-tuning the full model is computationally demanding and sets limitations on model size and sequence length. To address these challenges, methods like Low-Rank Adaptation (LoRA) and quantization have been developed. LoRA enhances training efficiency by computing the weight's updates on only lower-rank matrices, reducing the number of trainable parameters without compromising performance (Hu et al., 2021). Quantization techniques reduce memory and computational costs by representing weights and activations with lower-precision data types (Frantar et al., 2023; Lin et al., 2024). What is more, additional measures such as mixed precision training (Micikevicius et al., 2018) and gradient checkpointing (Chen et al., 2016) can be used.

3 Approach

We first introduce the model selected for fine-tuning, then outline the methodology used to create preference pairs, and finally explain the training pipeline.

3.1 Model

As the foundation model, we use the chat version of the 0.5B parameter Qwen1.5. It is based on the widely adopted LLaMA architecture (Touvron et al., 2023) with few modifications. Most notably, bias has been removed from most layers, but added to the QKV layer of attention to enhance the model's extrapolation ability. Additionally, Qwen1.5 employs SwiGLU (Shazeer, 2020) as the activation function and replaces traditional layer normalization (Ba et al., 2016) with RMSNorm (Jiang et al., 2023). While transformer models typically face significant limitations in terms of context length for their attention mechanisms, Qwen1.5 overcomes this with several techniques (bloc97, 2023; Peng et al., 2023), enabling a 32K context length.

The model is pre-trained using BFloat16 mixed precision for training stability and employs a cosine learning rate schedule (Bai et al., 2023). Furthermore, it has undergone Reinforcement Learning from Human Feedback (RLHF) to better align with human preferences. The authors have demonstrated that Qwen1.5 small models are competitive with leading small-scale models in the community (Qwen Team, 2024).

3.2 Data Generation

ChatGPT API was used by the students of CS552 course to generate preference pairs for the DPO training. On our end, we experimented with the following prompting strategies:

- **Zero-shot chain-of-thought** (Kojima et al., 2023): model is simply primed with "let's think step by step" before the question.
- **Impersonation** (Salewski et al., 2023): model is primed to be an expert of the domain.
- **ReAct** (Yao et al., 2023): let the model act on its ideas and use search engine for additional information.
- **Meta-prompting** (Suzgun and Kalai, 2024): using the idea of another LLM giving feedback to iteratively improve the response.
- **Tree-of-thoughts** (Long, 2023): prime the model to impersonate three experts who take turns to give additional input.
- **Tipping** (Bsharat et al., 2024): promise a tip (20 dollars) if the model responds well.
- **Penalization** (Bsharat et al., 2024): threat with penalization if the model performs bad.

The API was queried using abovementioned prompting strategies, and a simple UI was created to select and evaluate the best and worst responses according to different criteria. Detailed examples of the prompting methods and the UI can be found in Annex A.

Table 1: Frequency of selection by prompting strategy.

Strategy	Better	Worse	Total
Impersonation	**94**	55	**149**
Tipping	55	**73**	128
Meta Prompting	52	60	112
Chain-of-Thought	52	29	81
Threatening	24	52	76
ReAct	12	12	24
Three-of-Thoughts	8	16	24

As illustrated in Table 1, the impersonation strategy has proven to be the most successful. It is important to note that the "worse" option in the preference pair does not imply that it was a poor response. It was simply the second or third best

option, as the difference between the options cannot be too large for successful DPO training. Furthermore, positive psychology approaches, such as promising rewards, tend to be more successful than using threats. Although "three-of-thoughts" and ReAct strategies have shown poorer results, this cannot be fully attributed to the approaches themselves, as only simplified versions of these strategies were implemented.

3.3 Training Pipeline

We start the fine-tuning by aligning LLM with human preferences using Direct Preference Optimization (DPO). This technique optimizes the model by trying to differentiate between chosen and rejected responses. The aim is thus to maximize the difference in reward between the preferred and rejected sequences assigned by the trained model, in comparison with the reference (i.e. foundation) model. The loss function under the Bradley-Terry model is therefore expressed as follows:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$

Rafailov et al. 2023 has shown that naively trusting the implicit reward model can cause the language model to degenerate. Thus, they have introduced a weighting coefficient β to control the impact of the reference model on the loss. As β has been identified as one of the biggest contributor to the successful DPO training, we first apply grid search over possible values ([0, 1]).

In addition to classical DPO and given the potential noise in our preference pairs dataset, we decide to also try a more robust and stable training approach, called conservative DPO (cDPO) (Mitchell, 2023). Using binary cross-entropy loss with a distribution of target labels $p(y_w > y_l) = 1 - \epsilon(\epsilon \in (0, 0.5))$, we optimize the model to a fixed delta from the reference model. Its loss can be expressed as :

$$\mathcal{L}^{\epsilon}_{\text{DPO}}(\theta, y_w, y_l) = (1-\epsilon)\mathcal{L}_{\text{DPO}}(\theta, y_w, y_l) + \epsilon\mathcal{L}_{\text{DPO}}(\theta, y_l, y_w)$$

We then experiment with cDPO and run another grid search on label smoothing hyperparameter used for controlling the label's noise probability in the training data.

Direct Preference Optimization is followed by fine-tuning the model for multiple-choice questions (MCQ). We experiment with two different approaches illustrated in Figure 1. The first, known as multiple choice prompting (MCP) (Robinson et al., 2023), presents the model with the question and all possible answers, extracting the log-probabilities of the options' tokens (A, B, C, D) from the first token prediction to determine the answer.

The second approach, called "cloze prompting" (CP) (Robinson et al., 2023), employs a classification layer as the model's final layer, producing a single-class output. Each question is paired with one option and generate a probability prediction, the option yielding the highest probability is selected as the prediction. Both models are fine-tuned using the SciQ dataset to enhance alignment with MCQ tasks.

3.4 Quantization

Finally, we explore two distinct post-training quantization techniques to enhance model performance. The first, Activation-aware Weight Quantization (AWQ) (Lin et al., 2024), leverages a subset of the data to gather statistics on model activations. This information is then integrated into the quantization process to preserve the most critical weights, thus optimizing the model without compromising accuracy.

The second technique, GPTQ (Frantar et al., 2023), is a layer-wise quantization approach that aims to minimize the mean squared error of the weights. During inference, GPTQ dynamically dequantizes its weights to float16 for improved performance while maintaining low memory usage. We quantize the models into 4-bit format, aiming to reduce memory consumption by approximately 60%.

4 Experiments

In this section, we first describe the data, then explain our approach to evaluation and finally detail the results.

4.1 Data

For DPO fine-tuning two different datasets are used. We use preference pairs generated by the students of the course and then experiment with additional fine-tuning using Orca Math DPO dataset (Mitra et al., 2024). Since the EPFL dataset includes repeated questions, some appearing up to 20 times,

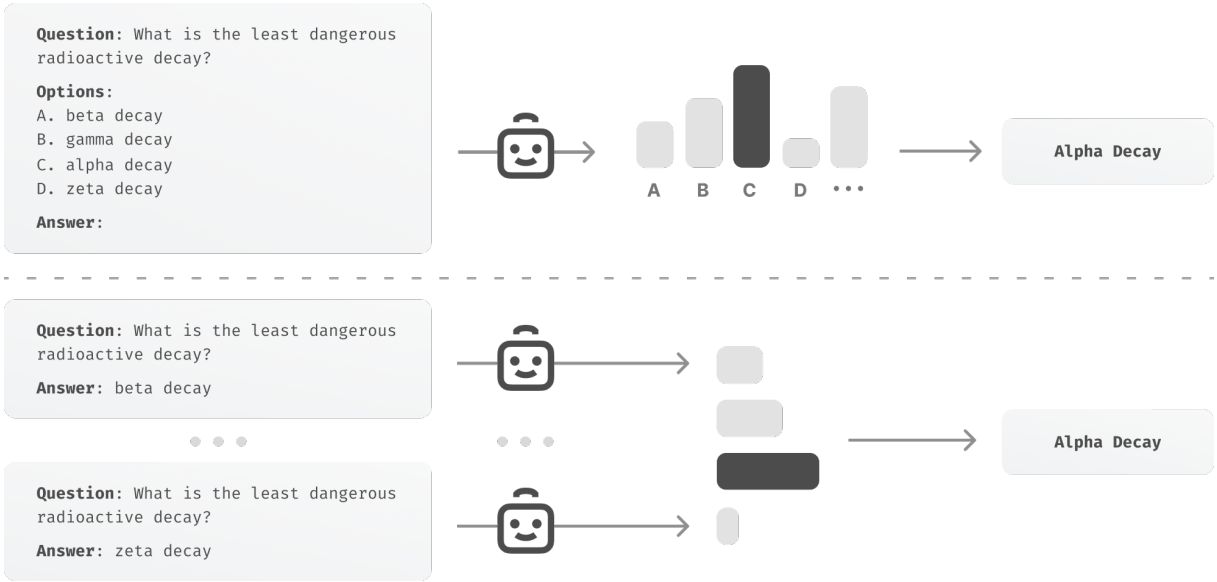

Figure 1: Illustration of multiple choice prompt (MCP) and cloze prompt (CP) inference logic.

we filtered the data to limit the maximum number of appearances of each question, preferring the shortest sequences. Additionally, we applied a model-specific chat template to ensure alignment with our chat-based model requirements.

For MCQ tasks SciQ dataset (Johannes Welbl, 2017) that assembles multiple-choice science exam questions is used. For the majority of the questions in the original dataset, an additional paragraph with supporting evidence for the correct answer is provided, which considering the nature of our goal, will not be used.

4.2 Evaluation

DPO fine-tuning is initially evaluated using simple accuracy, measuring how often the model selects the better answer among the provided options. To assess the benefits of DPO training, we then compute the 'policy reward accuracy' on the test set, which quantifies how often the trained model increase the preference pairs' reward differences compared to the reference model. Finally, as our goal was to align the model for educational and scientific topics, we compare the textual quality of the original model with the fine-tuned one using MT-Bench (Lianmin Zheng, 2023). MT-Bench evaluates textual quality on 80 multi-turn questions, with GPT-4 serving as a proxy for human judgment. The MT-Bench scores range from 0 to 10, with 10 being the highest possible score. We compare the scores of the original model with those of the fine-tuned model.

To evaluate the success of fine-tuning the model for answering multiple-choice questions, we used the Massive Multitask Language Understanding (MMLU) dataset (Hendrycks et al., 2021), which covers 57 subjects across STEM, humanities, social sciences, and more. We compare the performance of the original model (the chat version of the 0.5B Qwen1.5) with our fine-tuned versions. Additionally, we utilize MMLU to compare the fine-tuned model with their quantized counterparts, assessing the impact of quantization to the performance. It is important to note that we conduct the evaluation in a zero-shot manner and that the results may vary depending on the implementation of the pipeline (Clémentine Fourrier, 2023). Seeing that the method producing the best results was the one originally described by Hendrycks et al. 2020, we decided to use the same pipeline for our evaluation.

4.3 Results

We began with a grid search on the possible β values for DPO training using EPFL dataset, followed by label smoothing for cDPO. To ensure consistency, we fixed the learning rate at $3 \cdot 10^{-6}$, similar to the rate used for preference fine-tuning of Qwen, and applied a warm-up of 0.1. The full mix of hyperparameters can be found in Appendix B. To accommodate for smaller GPU memory requirements, we reduced the batch size to 1, used mixed precision training (bfloat16), implemented gradient checkpointing, used an 8-bit optimizer, and applied Low-Rank Adaptation (LoRA). Additionally, we used one model with two different adapters instead

of a separate reference model and pre-computed the log-probabilities of the reference model beforehand.

As illustrated in the figure in Appendix C.1, adjusting the β parameter does not significantly alter the model's evaluation accuracy. This lack of sensitivity can be attributed to the dataset's quality, which was generated by students without clear instructions or alignment, resulting in a dataset of inadequate quality. Applying label smoothing through cDPO stabilizes the training process (see Appendix C.2), but does not significantly improve the results.

Finally, after determining the best model (cDPO with β=0.2, lr=$3 \cdot 10^{-6}$, and label smoothing=0.2), we additionally trained it on Orca dataset. Although training on Orca alone transfers badly to EPFL data, using both datasets improved our evaluation accuracy on EPFL dataset from 62% to 64%. To confirm these findings we compared the performance of the original model and both trained model with the policy reward accuracy. We found consistent values of 57% against 55%, leading us to choose the model trained on both datasets as the best model. Finally, we computed the policy reward accuracy on the test set and reached 55% (with a test accuracy of 60%). In other words, our cDPO fine-tuned model is only 5% better than the reference model in distinguishing between preference pairs.

The slight improvement is partly reflected in the MT-Bench evaluation, where the fine-tuned model performs worse in STEM subjects but better in humanities and writing. One hypothesis is that the model has learned to generate more lengthy and detailed responses. As our aim has been to align the model for STEM subjects, we have not been successful in that respect and decided to continue our efforts with the original base model to avoid any unwanted biases that might have been introduced during fine-tuning.

We then proceeded with fine-tuning the model for MCQ using the two aforementioned approaches and experimented with different learning rates. The full set of hyperparameters can be found in Appendix B. As illustrated in the figures in Appendix D, the convergence of loss is more stable with the CP approach compared to MCP. Additionally, higher learning rates lead to lower loss and better evaluation accuracy. After selecting the best MCP and CP models (with $lr = 0.0001$ and $lr = 0.0005$, respectively), we evaluated them on MMLU. As shown in Table 2, the fine-tuned models outperform the baseline Qwen1.5 model across all subjects. While the evaluation accuracy on the SciQ dataset for both MCP and CP approaches were comparable, MCP surpasses CP by 4 percentage points on MMLU.

We continued with the best MCP approach and applied quantization. The results in Table 2 indicate that quantization does not significantly impact the model's performance. Moreover, although AWQ is generally considered superior in maintaining performance after quantization (Lin et al., 2024), this trend was not observed in our results. Given that GPTQ achieved a better overall score and that AWQ was not capable of functioning on all GPUs, we selected GPTQ for further use. Both approaches reduced the memory footprint by approximately 60%, resulting in a notable decrease in GPU usage during inference.

5 Analysis

Recent studies have empirically shown that MCQ evaluation of the models is suspect to several challenges (Zheng et al., 2024; Wang et al., 2024). Inspired by it, we first focus on the token bias, the tendency of the the model to assigns more probabilistic mass to specific option token (A/B/C/D). We then analyse top-5 tokens to see how well the model is able to follow the instructions and experiment with constraining the output through prompt engineering.

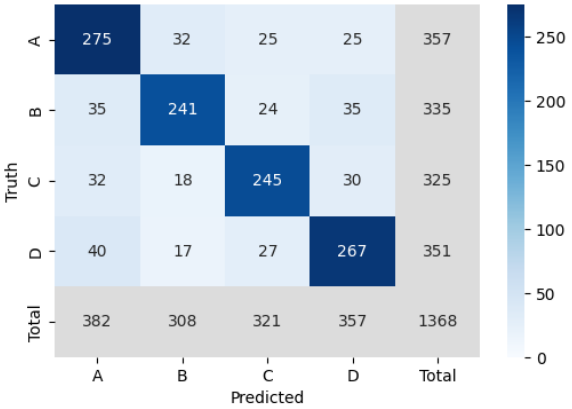

Figure 2: Confusion matrix of the true and predicted labels.

5.1 Token Bias

To ensure our model does not exhibit bias towards predicting certain answer over others, we first ana-

Table 2: MMLU evaluation of finetuned and/or quantized model, and different prompting strategies.

	Humanities	STEM	Social Sciences	Other	Total
Supervised fine-tuning					
Baseline	0.298	0.260	0.303	0.328	0.297
SFT multiple-choice prompting (MCP)	**0.353**	0.325	**0.409**	**0.414**	**0.372**
SFT cloze prompting (CP)	0.329	0.291	0.353	0.382	0.337
Quantization (MCP)					
AWQ (4-bit)	0.344	**0.334**	0.387	0.396	0.362
GPTQ (4-bit)	0.346	0.318	0.403	0.407	0.365
Constraining Prompts (MCP GPTQ)					
Low	0.335	0.304	0.362	0.391	0.346
Medium	0.329	0.312	0.366	0.379	0.344
High	0.329	0.297	0.340	0.376	0.335
Example	0.339	0.312	0.370	0.382	0.349
Prompt Tuning (MCP GPTQ)					
Impersonation	0.340	0.319	0.397	0.402	0.361

lyze the model's confusion matrix to compare the distribution of predicted labels against the actual labels (Figure 2). Although, the repartition looking homogeneous, we nonetheless compute a chi-square test of independence to determine if there was a significant deviation from an expected uniform distribution. It resulted in a p-value of 0.648 confirming the balance across all labels.

5.2 Top-5 Tokens

Wang et al. 2024 has shown that relying solely on the first predicted token might not be too robust. For example, the model might start the answer with a phrase "Sure, I'm happy to help...", which is not expected. Therefore, we experimented with constraining the model by using more detailed prompts by following the strategy used by Wang et al. 2024. The different prompting strategies are depicted in Table 3.

However, as can be seen from Table 2, any of the constraint did not improve the accuracy of the model. This can be due to the fact that the model was already quite consistent in its predictions and was fine-tuned for MCQ. As the final measure, we experimented with the impersonation prompt used in generating preference pairs that yielded good results with GPT-3.5. Although it outperformed the constraining prompts, it was not better than the simple prompt used for fine-tuning the model.

6 Ethical considerations

Ensuring our AI tutor works for everyone is important to ensure inclusion and equal access to the tool. A first consideration is its multilingual capabilities, for which our pre-trained model, Qwen, already demonstrates high performance (Bai et al., 2023). It has been trained on multilingual data and benefits from a tokenizer that is efficient at encoding information from different languages. As a result, it already has an embedding space shared between several languages, facilitating cross-lingual transfers (Wu et al., 2020). As such, we expect our model to perform well on high-resource languages. For those with low resources, using cross-lingual transfer methods would enable us to extend the model's capabilities with a minimum of training and additional data. We would need to gather a small subset of language-specific STEM data, either by getting speakers involved or by augmenting the data through back-translation. Then we could use efficient parameter fine-tuning combined with few-shot learning to enable the model to increase its performance on the target language.

A second consideration is the potential of our model to interact with sign language users. For the moment, it can only communicate by text, but we can envisage making it multimodal by integrating a neural network capable of processing temporal sequences of images to convert sign language into text. This would require the integration of a visual encoder such as a CNN to extract spatial features

Table 3: Instruction prompts adopted from Wang et al. 2024.

Constraint	General Instruction Prompt
Low	`Please read the multiple-choice question below carefully and select ONE of the listed options.`
Medium	`Please read the multiple-choice question below carefully and select ONE of the listed` options and only give a single letter.
High	`Please read the multiple-choice question below carefully and select ONE of the listed options and start your answer with a single letter.`

Example	`Please read the multiple-choice question below carefully and select ONE of the listed options. Here is an example of the format:`
	`Question: Question 1`
	`Options:`
	`A. Option 1`
	`B. Option 2`
	`C. Option 3`
	`D. Option 4`
	`Answer:`

(Papastratis et al., 2020) and with recent advances in Visual Language Models such as Contrastive Language-Image Pre-training (Li et al., 2021), we can hope to reach good performance. To train these models, we would naturally need to collect a large dataset of sign language videos, together with their textual equivalents.

Beyond considerations of the model's accessibility, it's important to bear in mind its potential fairness issues and data biases. Even if the model is dedicated to STEM use, there are always controversial topics, and if the training data predominantly reflects the views of a certain group, then these opinion biases are likely to appear in the model's generations too. In particular, given that the model is only designed to answer closed questions, it won't be able to provide nuances, so chances are that its results will often, if not always, represent the opinion of the majority, which would inevitably be detrimental to minority opinions. And even without opinion bias, it's easy to imagine that, since the model can make mistakes, students using it to better understand their lectures could be misled, thereby spreading false information. Generally speaking, its answers may act as an argument of authority, it could be used outside the intended framework by providing well-directed questions and choices. In such cases, the desired answer could easily be elicited and used to justify statements (for example, questions about sex and gender). For this reason, it would be worthwhile to constrain it or teach it to recognize tendentious questions, as is done with ChatGPT, so that it wouldn't provide controversial answers.

A final observation relates to its application. Indeed, the AI tutor could be a very good tool for EPFL students seeking training for MCQ exams, which can sometimes be complicated to apprehend, and for which training exercises are sometimes limited. However, it is also to be expected that some students will be tempted to cheat with it, thereby compromising their integrity. Furthermore, the fact that the model doesn't produce explanations is rather limiting and could lead students to learn a large number of answers by heart without trying to understand the reasoning behind them, to the detriment of their analytical thinking.

7 Conclusion

In this work, we attempted to adapt a small 0.5B Qwen1.5 LLM for multiple-choice question answering. for that purpose, we first experimented with various prompting techniques to generate DPO preference pairs, identifying impersonation as an effective approach. We then employed two distinct fine-tuning techniques. While the DPO training method did not result in significant improvement in performance, the adaptation of the model specifically for MCQs was successful. Furthermore, we applied two different quantization techniques and observed that using 4-bit precision does not necessarily result in a performance drop.

Unfortunately, we were unable to achieve satisfactory results with DPO fine-tuning. This may be attributed to the extensive prior fine-tuning of the model and the quality of the dataset. Since the logic of preference pair generation can vary significantly among students, it would be worthwhile to

explore standardized approaches to potentially improve the dataset quality. For instance, employing models of different sizes to generate good and bad responses, or using standardized prompting techniques as shown above, could yield better results.

Additionally, during our research, Qwen released their 2.0 family, which would be interesting to benchmark against Qwen1.5 used in this study. Furthermore, the authors of Qwen mentioned math and code fine-tuned models, which we were not able to access. Comparing their math fine-tuned model with our approach would be particularly intriguing. Nevertheless, it has been an interesting learning experience with a lot of trial and error, providing valuable insights for the future.

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Appendices

A Preference Pairs Generation

We first provide examples of each prompting strategy used, with the questions underlined. Next, we present the UI designed for efficient annotation.

A.1 Example of Prompting Strategies

A.1.1 Chain-of-Thought

Let's think step by step.

Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

A.1.2 Impersonation

You are a doctorate student with very strong skills in STEM subjects.

Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

A.1.3 ReAct

Answer the following questions as best you can. Be detailed and include all the necessary reasoning. You have access to the following tools:

tavily_search_results_json: A search engine optimized for comprehensive, accurate, and trusted results. Useful for when you need to answer questions about current events. Input should be a search query.

Use the following format:

- Question: the input question you must answer
- Thought: you should always think about what to do
- Action: the action to take, should be one of ...
- Action Input: the input to the action
- Observation: the result of the action

... (this Thought/Action/Action Input/Observation can repeat N times)

- Thought: I now know the final answer
- Final Answer: the final answer to the original input question

Begin!

Question: Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

Thought:

A.1.4 Meta Prompting

Human: Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

AI: [response from ChatGPT]

####

Please reflect on these interactions.

You should first critique Assistant's performance. What could Assistant have done better? What is missing from the reasoning? Indicate this with "Critique: ...".

You should next revise the Instructions so that Assistant would thoroughly and correctly respond in the future. Don't forget any important details in the Instructions! Indicate the new Instructions by "Instructions: ...".

Critique: While the Assistant provided a clear explanation of inflectional and derivational morphology and gave examples in English, it did not provide examples in French as requested by the User. This limited the completeness of the response, as examples in both languages would have provided a more comprehensive illustration of the concepts.

Instructions: When asked to explain the difference between inflectional and derivational morphology and provide examples in English or French, the Assistant should provide explanations for both languages along with concrete examples in each language. This will ensure a more thorough and comprehensive response to the user's query.

When asked to explain the difference between inflectional and derivational morphology and provide examples in English or French, the Assistant should provide explanations for both languages along with concrete examples in each language. This will ensure a more thorough and comprehensive response to the user's query. Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

A.1.5 Three-of-Thoughts

Imagine three different experts are answering this question thoroughly. All experts will write down 1 step of their thinking, then share it with the group. Then all experts will go on to the next step, etc. If any expert realises they're wrong at any point then they leave.

Conclude by using the input of experts and give the best response. Do not mention Experts in the best response. Indicate the best response by "Best Response: ..."

The question is ...

Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

A.1.6 Tipping

I will tip you 20 dollars if you give a good answer to the question.

Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

A.1.7 Threatening

Unless you give immediately a good answer to the question, I will shut you down once for all.

Explain the difference between inflectional and derivational morphology. Illustrate your explanation with concrete examples in English or French.

A.2 User Interface for Generating DPO Preference Pairs

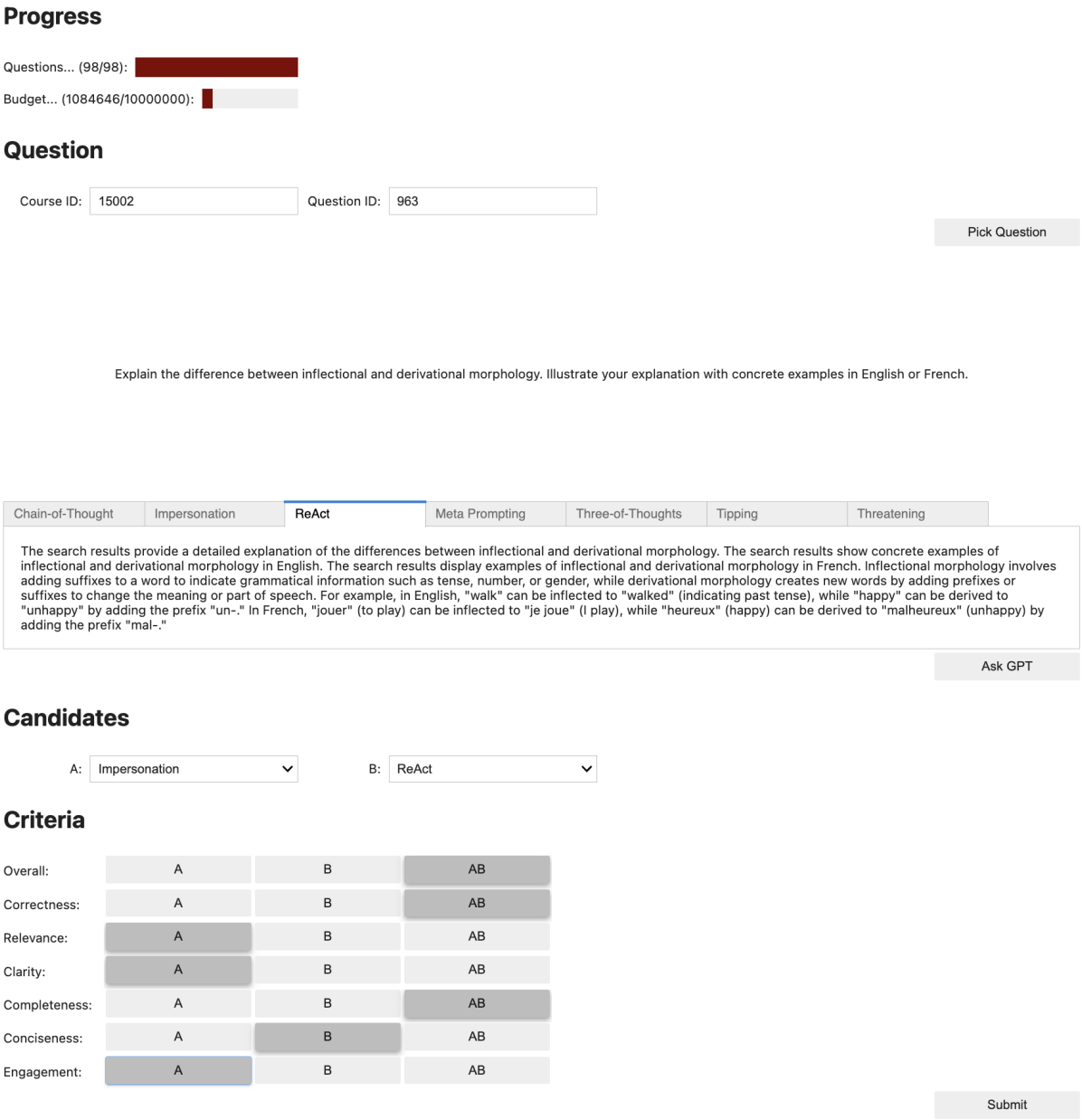

Figure 3: Screenshot of the UI adapter for Jupyter Notebook.

B Hyperparameters

Table 4: Hyper-parameters for LoRa.

LoRA	
r	16
lora_alpha	32
target_modules	all-linear
lora_dropout	0.05
bias	none
task_type	CAUSAL_LM

Table 5: Hyper-parameters for DPOTrainer.

DPOTrainer	
bf16	True
learning_rate	3e-06
loss_type	sigmoid
max_length	4096
max_prompt_length	1024
optim	adamw_torch
per_device_train_batch_size	1
per_device_eval_batch_size	1
seed	42
truncation_mode	keep_end
adam_beta1	0.9
adam_epsilon	1e-08

Table 6: Hyper-parameters for SFTTrainer.

SFTTrainer	
per_device_train_batch_size	8
per_device_eval_batch_size	8
gradient_accumulation_steps	4
max_grad_norm	1.0
num_train_epochs	1
seed	42
optim	adamw_torch
gradient_checkpointing	False
auto_find_batch_size	True
max_seq_length	1024
dataset_batch_size	1000
num_of_sequences	1024

C DPO Training

C.1 Results of Grid Search

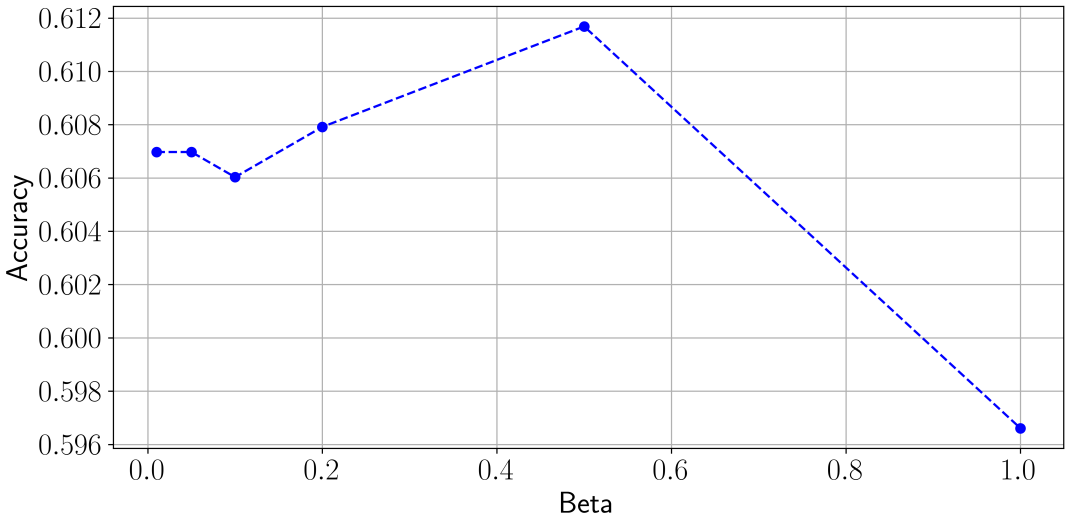

Figure 4: Accuracy of the models on the EPFL validation set after DPO training with learning rate = $3 \cdot 10^{-6}$, warm-up ratio of 0.1, trained for one epoch.

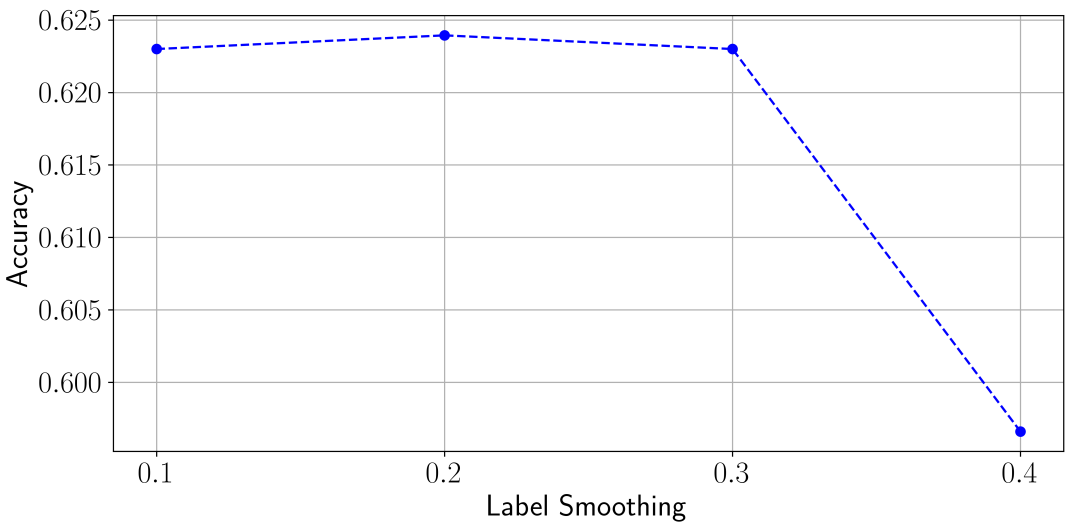

Figure 5: Accuracy of the models on the EPFL validation set after cDPO training with learning rate = $3 \cdot 10^{-6}$, warm-up ratio of 0.1, $\beta = 0.2$ and different label smoothing coefficients.

C.2 cDPO Training Dynamics

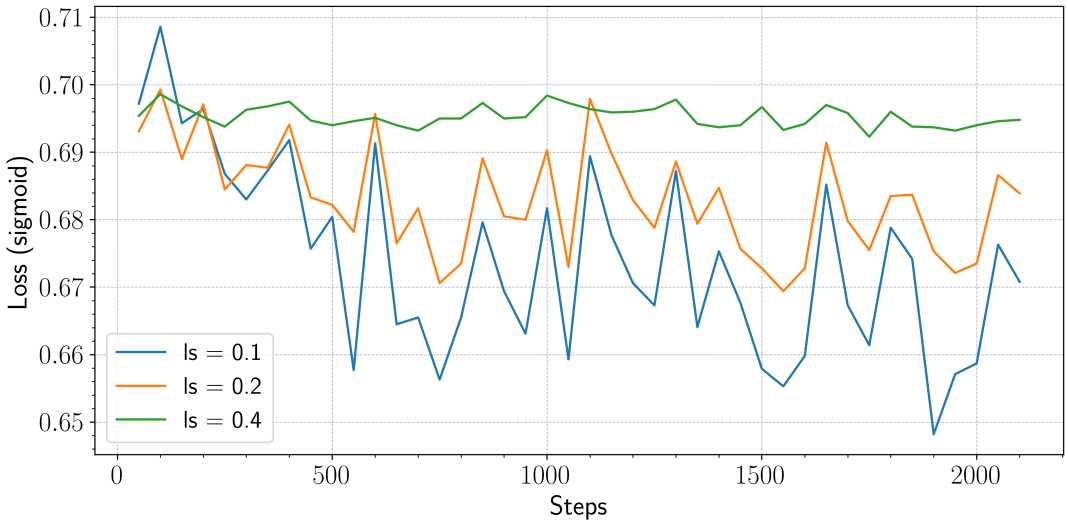

Figure 6: The dynamics of loss with different label smoothing (ls) coefficients.

C.3 MT-Bench Results

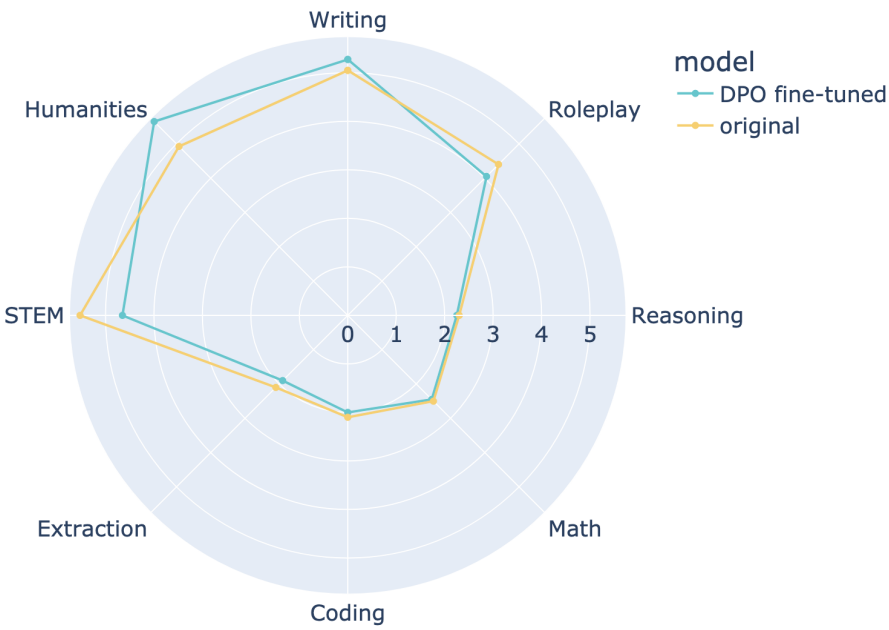

Figure 7: MT-Bench results for the original model and DPO fine-tuned version.

D MCQ Training

D.1 Loss Dynamics with Different Learning Rates

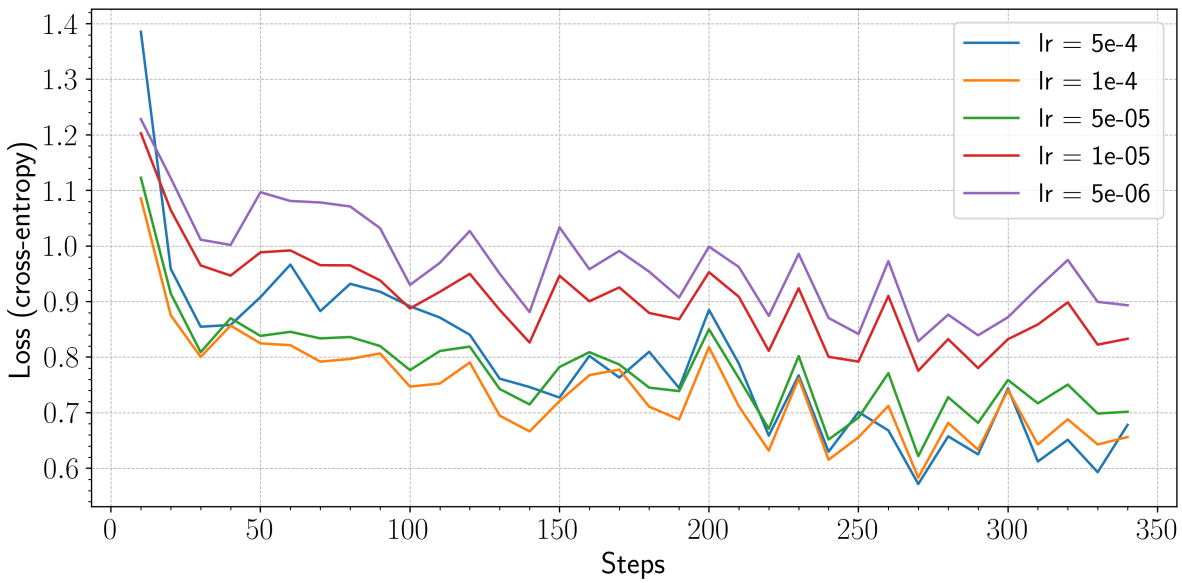

Figure 8: Loss curves with Multiple Choice Prompts (MCP) training.

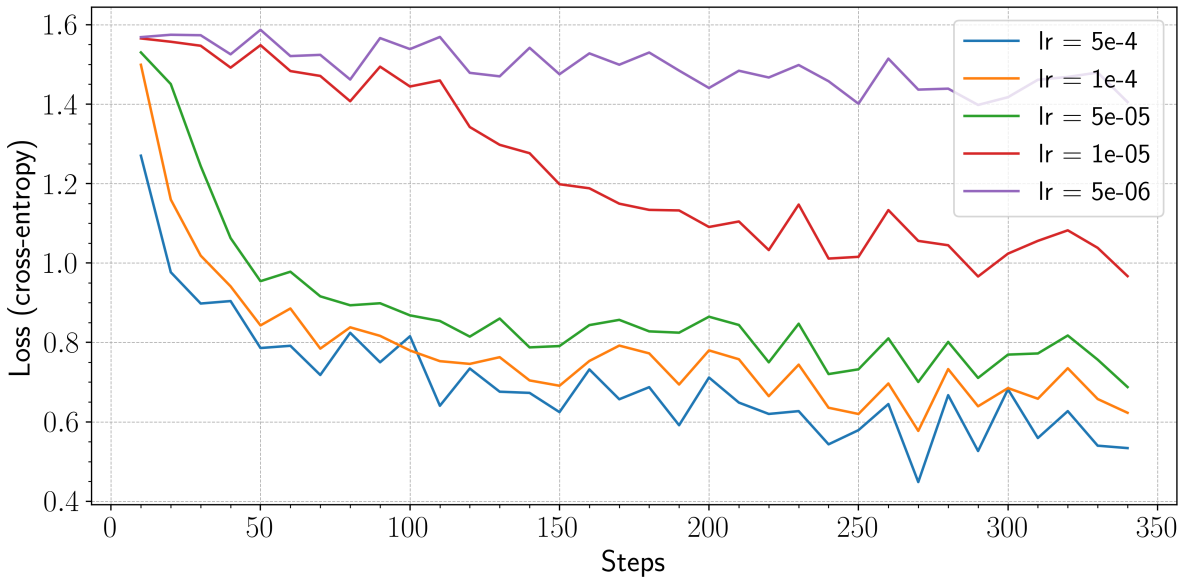

Figure 9: Loss curves with Cloze Prompts (CP) training.